

1. Define Spatial Filtering.
2. Describe averaging filters and the effect of mask size in image enhancement.
3. Explain why the median filter is better than the mean filter ?
4. Describe the characteristics and drawbacks of Ideal Low Pass Filter.
5. Generate the filtered image of the following image using a 3×3 averaging mask.(an image matrix will be given)Math
6. Describe suitable filtering techniques for restoring a medical X-ray image corrupted by Gaussian noise.
7. Develop a median filtering method for removing salt-and-pepper noise from a corrupted image.
8. Thresholding.
9. Basic global thresholding limitation
10. Design an adaptive thresholding technique to segment a grayscale image.(include diagram)
11. Define image segmentation.Its importance
12. build a region-based segmentation process for a given image.(just explain how the region based segmentation works step by step)
13. Define structuring element,erosion and dilation.
14. Describe the applications of boundary extraction in digital image processing.
15. Design an image compression model and explain its major components.
16. Different thresholding methods.
17. Types of data redundancy.

Syllabus: starting from **Spatial filtering** - end